

# Amin Boumerdassi

*Early career researcher* 

 Personal website |  Email |  GitHub |  LinkedIn

With a mix of **machine learning** and **Bayesian statistics**, I develop and apply approaches for cheaper and more performant **data analysis** in **gravitational wave astronomy**. My past works include estimating noise-related inference biases for **extreme mass ratio inspirals** with Fisher matrices, and experimental noise removal through **convolutional neural networks**.

## Education

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### PhD in Physics

Supervised by Dr Matt Edwards.

Thesis: “Bayesian Tools and Machine Learning Methods for Gravitational Wave Data Analysis of Extreme Mass-ratio Inspirals”

*University of Auckland*

2022 – 2026

### Integrated Master of Physics (1<sup>st</sup> class Hon.)

Supervised by Prof. Patrick Sutton.

Thesis: “Real-time Classification of Transient Gravitational-Wave Noise Using Convolutional Neural Networks”

*Cardiff University*

2018 – 2022

## Awards & Scholarships

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### University of Auckland Doctoral Scholarship Extension

*University of Auckland*

Dec 2025 - May 2026

### Doctoral Scholarship (Marsden fund)

*Royal Society Te Apārangi*

Dec 2022 - Dec 2025

### ESA Academy Student Sponsorship

Funded for travel to 15th International LISA Symposium.

*European Space Agency*

2024

### Highly Commended Student Presentation

Talk: “Investigating the impact of gravitational wave glitches on the parameter estimation of extreme mass ratio inspirals”.

*New Zealand Statistical Association*

Dec 2023

### Georges Lemaître award for “Contribution to Community Engagement”

Awarded by the School of Physics and Astronomy.

*Cardiff University*

Jun 2019

## Teaching Experience

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### Graduate Teaching Assistant

Tutored and/or marked courses:

Probability and its Applications (STATS 125), Data Analysis (STATS 201/208), Statistical Theory (STATS 210).

*University of Auckland*

2023 – 2025

### Demonstrator

Tutored course: Computational Physics (PX3143).

*Cardiff University*

2021 – 2022

## Admin Experience

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### Committee Member

Arranged and liaised with sponsors of the 2025 New Zealand Mathematics and Statistics Postgraduate Conference (NZMASP). *NZMASP*  
2024 – 2025

### Webmaster

Maintaining, and updating website for NZ Gravity research group. *NZ Gravity*  
2024 – 2025

### Meeting Chair

Organised weekly group meetings for NZ Gravity research group. *NZ Gravity*  
2023 – 2025

### Student Representative

Represented PhD students in the Department of Statistics. *University of Auckland*  
2022 – 2023

## Publications

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Boumerdassi, Amin, Matthew C. Edwards, Avi Vajpeyi, and Ollie Burke (2026). “First-time assessment of glitch-induced bias and uncertainty in inference of extreme mass ratio inspirals”. In: *Phys. Rev. D* 113 (10), p. 104044. DOI: 10.1103/9678-764y.

## Conference Presentations

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### NZ Gravity Spring 2025 Workshop

Talk: “Assessing glitch-induced bias and variability in EMRI parameter estimation” *Dunedin, New Zealand*  
Nov 2025

### NZ Gravity Autumn 2025 Workshop

Talk: “EMRIs and their role in GW science” *Auckland, New Zealand*  
Mar 2025

### New Zealand Statistical Association Conference

Talk: “Investigating the impact of gravitational wave glitches on the parameter estimation of extreme mass ratio inspirals” *Wellington, New Zealand*  
Dec 2024

### New Zealand Mathematics and Statistics Postgraduate Conference

Talk: “Investigating the impact of gravitational wave glitches on the parameter estimation of extreme mass ratio inspirals” *Christchurch, New Zealand*  
Nov 2024

### 15th International LISA Symposium

Poster: “Signal Detection of EMRIs with Convolutional Autoencoders” *Dublin, Ireland*  
Jul 2024

### NZ Gravity Summer 2024 Workshop

Talk: “Using Convolutional Autoencoders for Signal Detection of Extreme Mass Ratio Inspirals Detected by the LISA Mission” *Auckland, New Zealand*  
Feb 2024

### New Zealand Statistical Association Conference

Talk: “Using Convolutional Autoencoders for Signal Detection of Extreme Mass Ratio Inspirals Detected by the LISA Mission” *Christchurch, New Zealand*  
Dec 2023

New Zealand Mathematics and Statistics Postgraduate Conference

Talk: “Using Convolutional Autoencoders for Signal Detection of Extreme Mass Ratio Inspirals Detected by the LISA Mission”

Wellington, New Zealand

Nov 2023

Professional Memberships

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|-------------------------------------|------------------------------------------------------|
| OzGrav                              | <div>Affiliate</div> <div>Jan 2024 - Present</div>   |
| New Zealand Statistical Association | <div>Member</div> <div>Sep 2023 - Present</div>      |
| LISA Consortium                     | <div>Core member</div> <div>Dec 2022 - Present</div> |
| NZ Gravity                          | <div>Member</div> <div>Dec 2022 - Present</div>      |
| LIGO Scientific Collaboration       | <div>Member</div> <div>Sep 2020 - Jun 2023</div>     |
| Institute of Physics                | <div>Member</div> <div>Oct 2019 - Present</div>      |

Referees

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| <div>Dr Matt Edwards</div> <div>Department of Statistics</div> <div>University of Auckland</div> <div>matt.edwards@auckland.ac.nz</div> | <div>Prof. Renate Meyer</div> <div>Department of Statistics</div> <div>University of Auckland</div> <div>renate.meyer@auckland.ac.nz</div> | <div>Dr Avi Vajpeyi</div> <div>Department of Statistics</div> <div>University of Auckland</div> <div>avi.vajpeyi@auckland.ac.nz</div> |
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